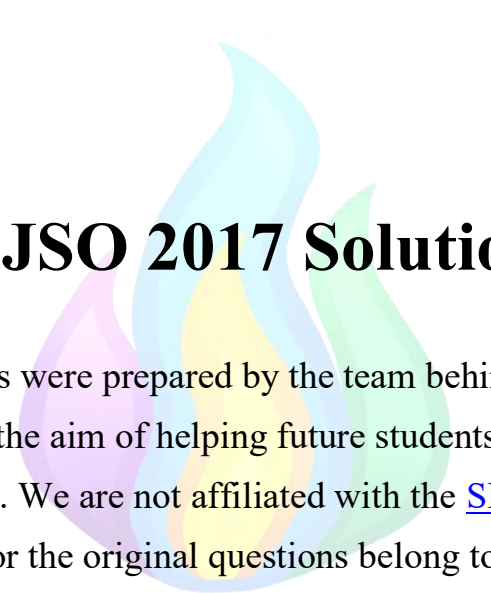




SLJSO 2017 Solutions

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<u>2</u>	Evolutionary Development	<u>Animal Classification</u>
<u>3</u>	Cell Biology	<u>Biomolecules</u>
<u>4</u>	Molecular Biology	<u>DNA and RNA</u>
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Question 1

Options (A) through (C) are incorrect, as prokaryotic cells have no membrane-bound organelles. Ribosomes, however, are found in all types of cells. Therefore, **option (D)** is the correct option.

Correct Answer: D

Question 2

Binomial nomenclature is typically underlined. Therefore, **option (D)** is correct.

Correct Answer: D

Question 3

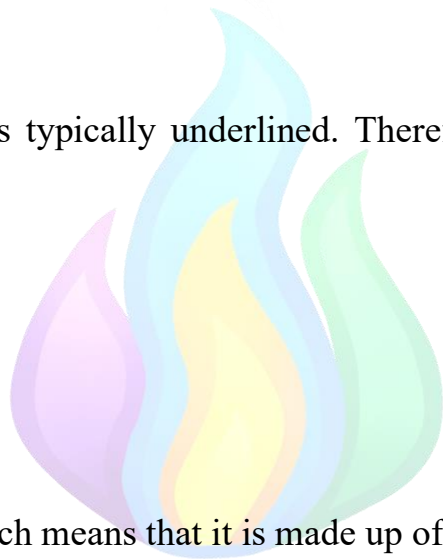
Albumin is a protein, which means that it is made up of **option (C), amino acids**.

Correct Answer: C

Question 4

DNA is known for having thymine, and RNA replaces thymine with uracil. Options (B) through (D) all have either of these. **Option (A)** has neither of those, so it can be found in BOTH DNA and RNA.

Correct Answer: A



Question 5

Option (A) is incorrect, as the trachea is lined with pseudostratified epithelium. Option (B) is incorrect because blood vessels are lined with simple squamous epithelium. Option (C) is incorrect; the inner wall of the Bowman's capsule is lined with podocytes, not simple columnar epithelium. **Option (D)** is correct—the wall of the PCT is lined with simple cuboidal epithelium, increasing its surface area to allow maximum reabsorption of water and electrolytes.

Correct Answer: D

Question 6

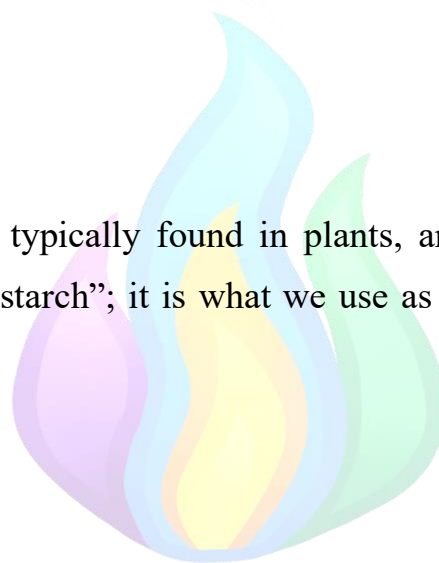
Cellulose and pectin are typically found in plants, and chitin is found in fungi. Glycogen is the “animal starch”; it is what we use as our energy source. Humans cannot digest cellulose.

Correct Answer: B

Question 7

Endemic organisms are those that are only found in and native to one place in the world, in this case Sri Lanka. Indigenous species are those that naturally occur in an environment and did not get introduced from another environment. Exotic species were introduced from another environment, the opposite of indigenous species. **Option (D)** is correct—the hora is endemic to Sri Lanka, the snakehead fish is indigenous to Sri Lanka, and the tilapia is an exotic species (introduced from Africa). The other options are either in the wrong order or don't match the correct organisms to their correct classification.

Correct Answer: D



Question 8

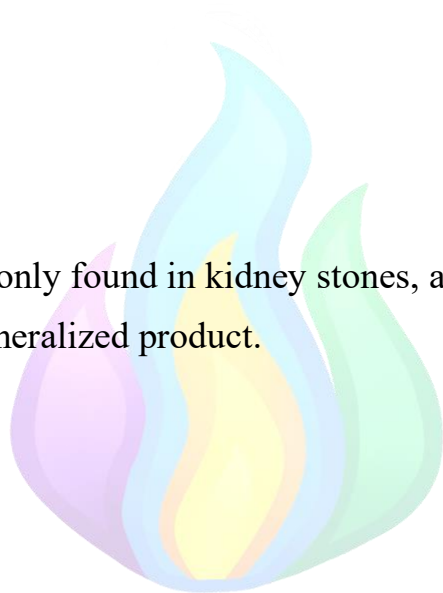
Petri dishes are typically sterilized using dry heat (often done in a hot air oven). Controlling microbes in water is done by filtration. Culture media is sterilized with moist heat. Milk is sterilized with pasteurization (low heat to kill pathogens). Therefore, the correct answer is **option (D)**.

Correct Answer: D

Question 9

Calcium oxalate is commonly found in kidney stones, as the calcium in urine reacts with oxalate to form a mineralized product.

Correct Answer: D



Question 10

Trypsin, **option (B)**, is the enzyme that breaks down proteins. Lipase breaks down fats, amylase breaks down carbohydrates, and cellulase breaks down cellulose.

Correct Answer: B

Question 11

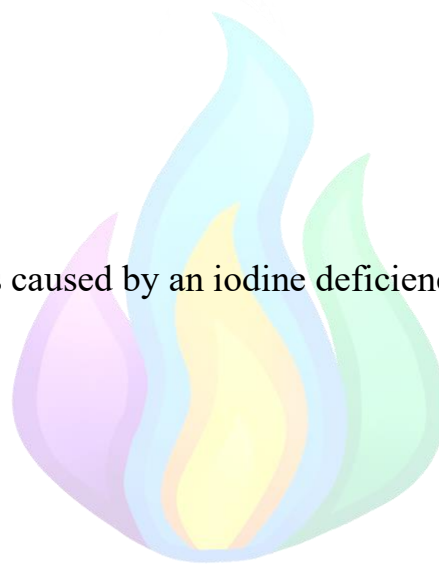
Option (A) is correct because Platyhelminthes (also called flatworms—they can be free-living or parasites) have functional flame cells. Option (B) is also correct because chordates (including humans and fish—kingdom Animalia) have a ventrally (anteriorly) placed heart relative to the nerve cord, meaning the heart is in front of the nerve cord. **Option (C)** is incorrect. Annelids have a closed circulatory system. Option (D) is correct: mollusks (snails, clams, etc.) have a muscular foot.

Correct Answer: C

Question 12

Simple goiter is an illness caused by an iodine deficiency option (A).

Correct Answer: A



Question 13

Color blindness is known to be an X-linked recessive trait. If the mother is colorblind, it must mean that she is homozygous recessive for this trait. As it is X-linked, the daughters will be carriers as they only inherit one recessive allele from their mother and a dominant allele from their father. The sons, however, will be colorblind because they inherit everything on the one X chromosome from the mother. The Y chromosome does not affect this trait. Therefore, **option (C)** is correct.

Correct Answer: C

Question 14

Nitrogen fixation, converting nitrogen gas into ammonium, is accomplished by Anabaena bacteria. Nitrification, the process of converting ammonium to nitrites and nitrates, is performed by Nitrosomonas (nitrite → nitrate) and Nitrobacter (nitrite → nitrate). Denitrification, the process of converting nitrate to nitrogen gas is accomplished by Pseudomonas. Therefore, **option (B)** is the correct answer. As for the other bacteria, clostridium is found in the intestinal tract of animals and humans. Thiobacillus oxidizes sulfur, producing sulfates for plants, not helping with the nitrogen cycle.

Correct Answer: B

Question 15

This question is testing knowledge of innate, acquired, active, and passive immunity. A is incorrect because infants receive antibodies from the mother's breast milk passively. For this to be active, the infant would have to fall sick and develop the antibodies firsthand. B is also incorrect because having the infection firsthand (having chickenpox) is an active immune process, not a passive one. D is correct because polio is artificially acquired in the form of a vaccine. It represents active immunity because the individual fights off the infection first-hand. D is unfortunately technically flawed, as receiving the infection in the form of a vaccine is artificially acquired **active** immunity (as the body must fight it firsthand). **C** seems to be the best option.

Correct Answer: C

Question 16

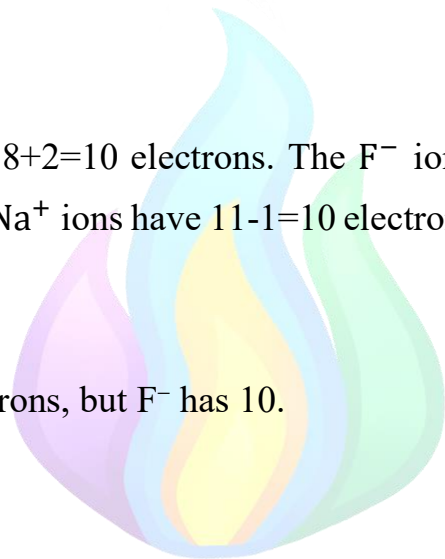
(A) False. An atom has an equal number of protons and electrons. But negative ions have gained electrons. So, they have more electrons than protons. On the other hand, positive ions have lost electrons. So, they have fewer electrons than protons. So, clearly the statement is wrong for Na ion and Mg ion.

(B) False. The number of protons and neutrons does not change when an atom becomes an ion. Oxygen has 8 protons and 8 neutrons. So, that statement is clearly wrong.

(C) True. O^{2-} ions have $8+2=10$ electrons. The F^{-} ion has $9+1=10$ electrons. Ne atoms have 10 electrons. Na^{+} ions have $11-1=10$ electrons. Mg^{2+} ions have $12-2=10$ electrons.

(D) False. O^{2-} has 8 neutrons, but F^{-} has 10.

Correct Answer: C



Question 17

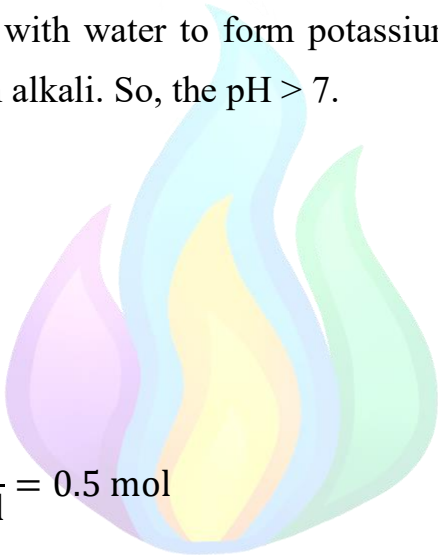
1 - True. Potassium is a metal. Metals have free electrons in the solid state. So, they conduct electricity in the solid state. In a molten state, metals have freely moved electrons and ions, which also conduct electricity. So, potassium conducts electricity in both states.

2 - Yes. Potassium is a very reactive metal, and it reacts explosively with water.

3 - No. Potassium reacts with water to form potassium hydroxide and hydrogen. Potassium hydroxide is an alkali. So, the $\text{pH} > 7$.

Correct Answer: A

Question 18


$$\text{Moles} = \frac{m}{M} = \frac{9\text{g}}{18\text{g/mol}} = 0.5 \text{ mol}$$

Then, the number of moles should be 0.5 mol.

$$\text{For option A, } n = \frac{m}{M} = \frac{4\text{g}}{2\text{g/mol}} = 2 \text{ mol}$$

$$\text{For option B, } n = \frac{m}{M} = \frac{28\text{g}}{28\text{g/mol}} = 1 \text{ mol}$$

$$\text{For option C, } n = \frac{m}{M} = \frac{16\text{g}}{32\text{g/mol}} = 0.5 \text{ mol}$$

$$\text{For option D, } n = \frac{m}{M} = \frac{44\text{g}}{44\text{g/mol}} = 1 \text{ mol}$$

Correct Answer: C

Question 19

A covalent bond is formed when electrons are shared between two nonmetals.

Correct Answer: B

Question 20

Since the number of Cu atoms should be the same, $p = r$.

Since the number of N atoms should be the same, $q = 2r + s$.

Since the number of H atoms should be the same, $q = 2t$.

Since the number of O atoms should be the same, $3q = 6r + s + t$.

The only choice that fits the above conditions is choice D.

Correct Answer: D

Question 21

To separate copper (II) sulfate crystals from sand, the correct sequence of processes is

Dissolving—Copper (II) sulfate dissolves in water, while sand does not.

Filtering—The sand is separated from the copper (II) sulfate solution via filtration.

Evaporating—The water is partially evaporated to concentrate the solution.

Crystallizing—Upon cooling, copper (II) sulfate crystals form.

Correct Answer: D

Question 22

- (A)- $\text{HCl} + \text{AgNO}_3 \Rightarrow \text{AgCl} \downarrow + \text{HNO}_3$
(B)- $\text{HCl} + \text{NaNO}_3 \Rightarrow \text{NaCl} + \text{HNO}_3$
(C)- $2\text{NaCl} + \text{Pb}(\text{NO}_3)_2 \Rightarrow \text{PbCl}_2 \downarrow + 2\text{NaNO}_3$
(D) $\text{NaCl} + \text{AgNO}_3 \Rightarrow \text{AgCl} \downarrow + \text{NaNO}_3$

Option B is the answer since every other option gives a precipitate.

Correct Answer: B

Question 23

- (A) - The oxidation state of O atoms in oxygen changes from 0 to -2.
(B) - The oxidation state of Cu and Zn changes.
(C) - The oxidation state of any atom does not change.
(D) - The oxidation state of Zn and H changes.

Correct Answer: C

Question 24

When 1 mol of HCl reacts with 1 mol of NaOH, 54 kJ of energy is released. So, when 0.1 mol of HCl reacts with 0.1 mol of NaOH, 5.4 kJ of energy is released.

Correct Answer: C

Question 25

Magnesium is the strongest reducing agent (most negative voltage), meaning it readily loses electrons and acts as the anode. Silver is the strongest oxidizing agent (most positive voltage), meaning it readily gains electrons and acts as the cathode. The large potential difference between Mg and Ag results in the highest possible voltage for this setup.

Correct Answer: B

Question 26

(A) When you increase the concentration of the acid, the rate of collisions increases. So, the rate of the reaction increases.

(B) When you increase the particle size of the zinc, the total surface area decreases, and the rate of collisions decreases. So, the rate of the reaction decreases.

(C) Increasing the pressure has no effect on the system because it does not affect the rate of the reaction. (However, if one of the reactions was a gas, then increasing the pressure would have increased the rate of collisions. But in this case, it has no effect.)

(D) Increasing the temperature would increase the kinetic energy of the particles, thus increasing the percentage of collisions having more than the activation energy to react. Increasing the temperature will also increase the rate of collisions as the particles now move faster. So, the rate of the reaction increases due to both reasons.

Correct Answer: B

Question 27

$$\text{Moles of KOH} = C V = 1\text{M} \times \frac{15}{1000}\text{L} = 0.015\text{ mol}$$

Since the ratio of HCl : KOH is 1:1,

$$\text{Moles of HCl} = \text{Moles of KOH} = 0.015\text{ mol}$$

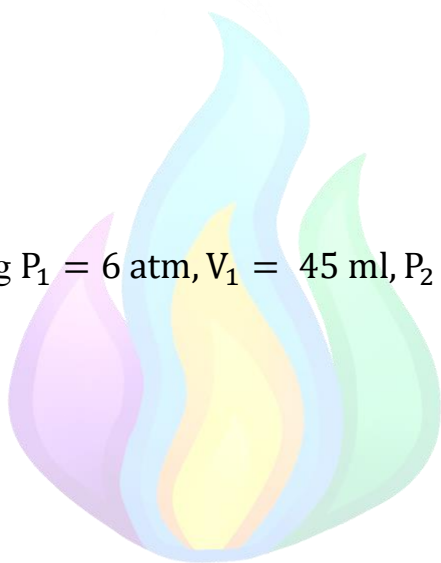
$$C = \frac{\text{Moles}}{V} = \frac{0.015\text{ mol}}{(20/1000)\text{L}} = 0.75\text{M}$$

Correct Answer: - A

Question 28

$$P_1 V_1 = P_2 V_2. \text{ Substituting } P_1 = 6\text{ atm}, V_1 = 45\text{ ml}, P_2 = 8\text{ atm}, \text{ we get}$$
$$V_2 = 33.75\text{ mL}$$

Correct Answer: B



Question 29

(A) Ammonia is a weak base and does not significantly react with either Na_2CO_3 or NaCl in a way that helps distinguish them.

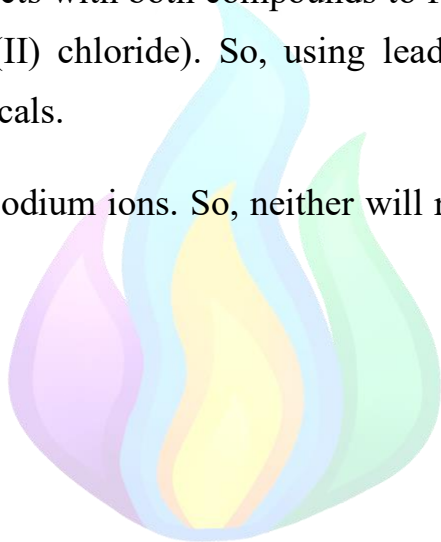
(B) HCl reacts with sodium carbonate to produce carbon dioxide gas.
$$\text{Na}_2\text{CO}_3 + 2\text{HCl} \rightarrow 2\text{NaCl} + \text{CO}_2 + \text{H}_2\text{O}$$

However, HCl does not react with NaCl . So, we can use this option to distinguish the two chemicals.

(C) - Lead (II) Nitrate reacts with both compounds to form white precipitates (lead (II) carbonate and lead (II) chloride). So, using lead (II) nitrate isn't useful to distinguish the two chemicals.

(D) - All reactions have sodium ions. So, neither will react in a way useful for us.

Correct Answer: B



Question 30

We can use Newton's second law to find the acceleration of an object:

$$F = m \times a$$

$$a = \frac{F}{m}$$

Since F and m are constants, that would mean that a must be a constant too.

Since the acceleration is constant, that means that the velocity, and with it, the momentum and kinetic energy, are all increasing.

Correct Answer: D

Question 31

Since this question uses mirrors, we can use the following formula:

$$\frac{1}{f} = \frac{1}{p} + \frac{1}{l}$$

Where the f is the focal length of the mirror, p is the distance from the object to the mirror, and l is the distance between the mirror and the image.

Since the mirror is convex, that means that the focal length and the image are both behind the mirror, because of which, we need to make these negative in the formula, giving us the following:

$$-\frac{1}{f} = \frac{1}{p} - \frac{1}{l}$$

$$\frac{1}{p} = \frac{1}{l} - \frac{1}{f}$$

Substituting l and f with the information given in the problem, we get

$$\frac{1}{p} = \frac{1}{10\text{cm}} - \frac{1}{15\text{cm}}$$

$$\frac{1}{p} = \frac{3}{30\text{cm}} - \frac{2}{30\text{cm}}$$

$$\frac{1}{p} = \frac{1}{30\text{cm}}$$

$$p = 30\text{cm}$$

Correct Answer: A

Question 32

Since the object gets all given energy, we know that

$$E = \frac{m \times \left(10 \frac{\text{m}}{\text{s}}\right)^2}{2} - \frac{m \times \left(0 \frac{\text{m}}{\text{s}}\right)^2}{2}$$

$$E = \frac{m}{2} \times \left(100 \frac{\text{m}^2}{\text{s}^2}\right)$$

We can similarly find the energy needed to accelerate the object from the velocity of 10 m/s to the velocity of 20 m/s:

$$E_1 = \frac{m \times \left(20 \frac{\text{m}}{\text{s}}\right)^2}{2} - \frac{m \times \left(10 \frac{\text{m}}{\text{s}}\right)^2}{2}$$

$$E_1 = \frac{m}{2} \times \left(\left(20 \frac{\text{m}}{\text{s}}\right)^2 - \left(10 \frac{\text{m}}{\text{s}}\right)^2 \right)$$

$$E_1 = \frac{m}{2} \times \left(400 \frac{\text{m}^2}{\text{s}^2} - 100 \frac{\text{m}^2}{\text{s}^2} \right)$$

$$E_1 = \frac{m}{2} \times \left(300 \frac{\text{m}^2}{\text{s}^2} \right)$$

Dividing E_1 by E we get

$$\frac{E_1}{E} = \frac{\frac{m}{2} \times \left(300 \frac{\text{m}^2}{\text{s}^2} \right)}{\frac{m}{2} \times \left(100 \frac{\text{m}^2}{\text{s}^2} \right)}$$

$$\frac{E_1}{E} = 3 \rightarrow E_1 = 3E$$

Correct Answer: C

Question 33

The problem involves using energy to increase temperature, for which it is obvious that we need to use the next formula to solve it:

$$Q = mc\Delta t$$

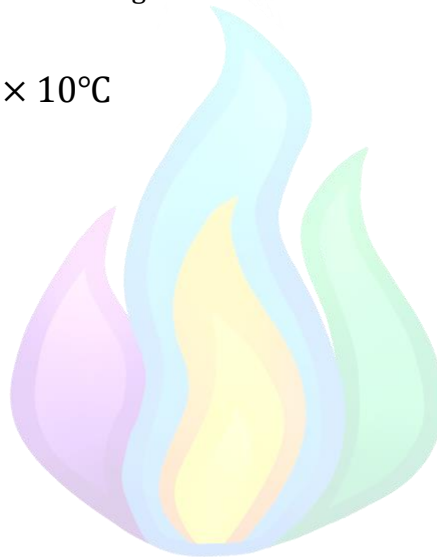
Q is the energy we need to calculate.

Substituting $m = 2\text{kg}$, $c = 4200 \frac{\text{J}}{\text{kg} \times ^\circ\text{C}}$ and $\Delta t = 10^\circ\text{C}$ we get:

$$Q = 2\text{kg} \times 4200 \frac{\text{J}}{\text{kg} \times ^\circ\text{C}} \times 10^\circ\text{C}$$

$$Q = 84000\text{J}$$

Correct Answer: A



Question 34

Condensers are used to transform steam into liquid. Ethanol easily evaporates, so the condenser would stop the ethanol gas from escaping by turning it back into liquid form, so it is used to stop it from escaping.

Correct Answer: B

Question 35

Let's say the mass of both liquids is m , then

$$m = \rho_1 V_1 = \rho_2 V_2$$

Where index 1 is used for water and index 2 is used for the liquid

$$1 \frac{\text{g}}{\text{cm}^3} V_1 = 2 \frac{\text{g}}{\text{cm}^3} V_2$$

$$V_1 = 2V_2$$

The overall volume of the mixture is

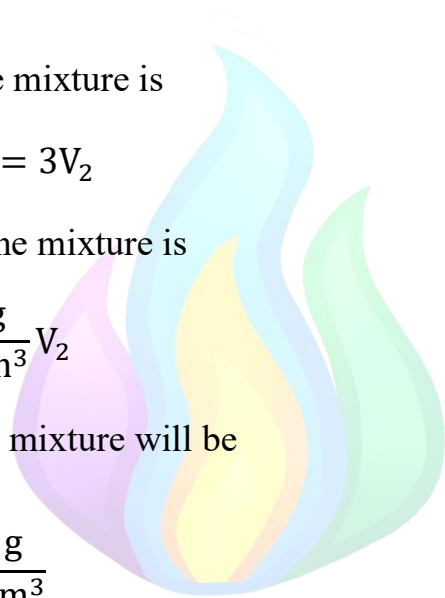
$$V = V_1 + V_2 = 2V_2 + V_2 = 3V_2$$

And the overall mass of the mixture is

$$M = 2m = 2\rho_2 V_2 = 4 \frac{\text{g}}{\text{cm}^3} V_2$$

The overall density of the mixture will be

$$\rho = \frac{M}{V} = \frac{4 \frac{\text{g}}{\text{cm}^3} V_2}{3V_2} = \frac{4}{3} \frac{\text{g}}{\text{cm}^3}$$



Correct Answer: B

Question 36

Since the lamp is working with the switch not being connected, once the switch is turned on, it is obvious that it will behave differently. Since there is no resistance in the part of the circuit with the switch, we can find the equivalent resistance for it using the next formula, which is used to find the equivalent resistance when resistors are connected parallel to each other:

$$\frac{1}{R_e} = \frac{1}{R_1} + \frac{1}{R_2}$$

So, since there is no resistance in the circuit in the part where the switch is, $R_2 = 0\Omega$

$$\frac{1}{R_e} = \frac{1}{R_1} + \frac{1}{0\Omega}$$

$$\frac{1}{R_e} = \infty \Rightarrow R_e = 0$$

Since there is no resistance, turning the switch on will increase the current, resulting in the lamp glowing brighter.

Correct Answer: A

Question 37

The only forces acting upon the object with the mass of $m = 3\text{kg}$ is the gravitational force and the tension, so we can find the acceleration of both objects (they are moving the same way because the string is tightened and has tension) using Newton's second law:

$$F = ma$$

$$F_t - F_g = ma$$

$$a = \frac{F_t - F_g}{m}$$

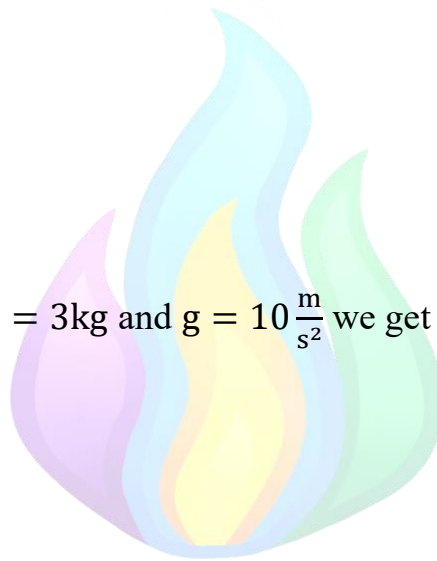
$$a = \frac{F_t - mg}{m}$$

Substituting $F_t = 39\text{N}$, $m = 3\text{kg}$ and $g = 10 \frac{\text{m}}{\text{s}^2}$ we get

$$a = \frac{39\text{N} - 30\text{N}}{3\text{kg}}$$

$$a = \frac{9\text{N}}{3\text{kg}}$$

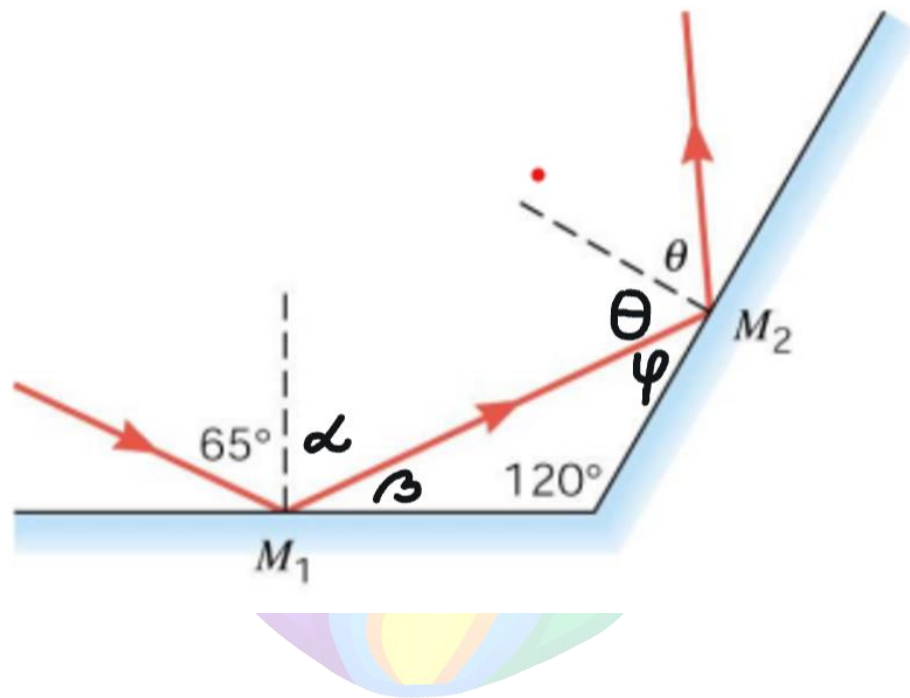
$$a = 3 \frac{\text{m}}{\text{s}^2}$$



Correct Answer: B

Question 38

Since the light is hitting a plane mirror, the angle at which the beam hits the mirror is equal to the angle at which it is leaving.



Because of this, we know that $\alpha = 65^\circ$. Since angles α and β are complementary, we get

$$\alpha + \beta = 90^\circ$$

$$\beta = 90^\circ - \alpha$$

$$\beta = 90^\circ - 65^\circ$$

$$\beta = 25^\circ$$

Since the triangle in the picture has the angles β , ϕ and 120° , the sum of them is 180°

$$\beta + \varphi + 120^\circ = 180^\circ$$

Substituting $\beta = 25^\circ$ we get

$$\varphi = 180^\circ - 120^\circ - 25^\circ$$

$$\varphi = 35^\circ$$

Since angles φ and θ are complementary, we get

$$\varphi + \theta = 90^\circ$$

$$\theta = 90^\circ - \varphi$$

$$\theta = 90^\circ - 35^\circ$$

$$\theta = 55^\circ$$

Correct Answer: D



Question 39

Looking at the graph, we see that the object travelled to a point 16m to the right from the starting point and then travelled back to the point that is 8m to the right from the starting point, so the body travelled $16\text{m} - 8\text{m} = 8\text{m}$ to get back to that point. The overall distance traveled is

$$s = 16\text{m} + (16\text{m} - 8\text{m}) = 16\text{m} + 8\text{m} = 24\text{m}$$

Correct Answer: C

Question 40

Since the rod is negatively charged, it will distribute its negative charge to the electroscope, leaving it negatively charged after we remove the rod.

Correct Answer: B

Question 41

We should write both sides using powers of 2.

$$\text{right hand side} = 2^{3x-1} * 4 = 2^{3x-1} * 2^2 = 2^{3x+1} \quad (X^a * X^b = X^{a+b})$$

$$\text{left hand side} = 2^{2x+2}$$

Since the left-hand side and the right-hand side are the same, $2^{3x+1} = 2^{2x+2}$,
 $3x + 1 = 2x + 2$. $X = 1$.

Correct Answer: C

Question 42

$$v^2 = u^2 + 2as$$

$$(v^2 - 2as)^{\frac{1}{2}} = u$$

Correct Answer: B

Question 43

A transverse wave, by its definition, has its particles moving perpendicular to their motionless state, which, in this case, falls on the line CI.

Correct Answer: C

Question 44

Pressure, for a solid object, is defined as the overall force acting on a certain area, divided by that same area.

$$p = \frac{F}{A}$$

The force in the formula is the weight of the liquid.

$$F = mg$$

The mass of the liquid is found by multiplying its density by the volume it takes up.

$$m = \rho V$$

$$p = \frac{\rho Vg}{A}$$

And this is the value of the starting pressure.

Let's calculate the pressure for every one of the given answers:

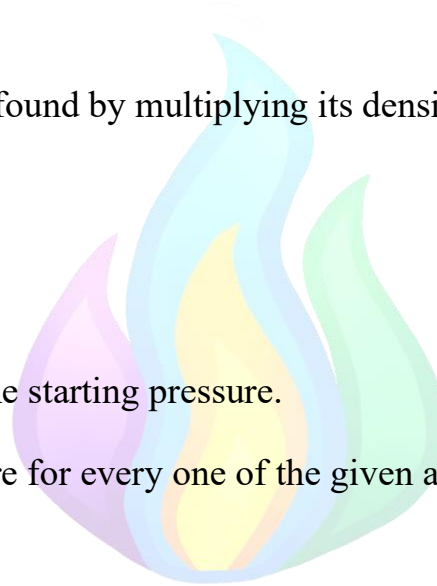
$$p_a = \frac{2\rho Vg}{\frac{A}{2}} = 4 \frac{\rho Vg}{A}$$

$$p_b = \frac{\frac{\rho}{2} 2Vg}{\frac{A}{2}} = 2 \frac{\rho Vg}{A}$$

$$p_c = \frac{2\rho Vg}{2A} = \frac{\rho Vg}{A}$$

$$p_d = \frac{\frac{\rho}{2} 2Vg}{2A} = \frac{1}{2} \frac{\rho Vg}{A}$$

Out of these 4, only answer choice C matches with the starting pressure.



Correct Answer: C

Question 45

Since the light ray comes back at the same trajectory it came from, we can rewrite this as a problem where we need to find the focal length of the mirror, where, if we place an image at the distance d from the mirror, its image is formed at the same distance. Using the mirror formula, we get

$$\frac{1}{f} = \frac{1}{d} + \frac{1}{d}$$

$$\frac{1}{f} = \frac{2}{d}$$

$$2f = d$$

$$f = \frac{d}{2}$$



Correct Answer: C

Question 46

When two lines are parallel, their gradient is similar. The gradient of the first line is 6. The gradient of the second line should also be 6. The y-intercept is said to be 8. So, the equation of the line should follow the following set of equations, where n is any integer.

$$n(y = 6x + 8)$$

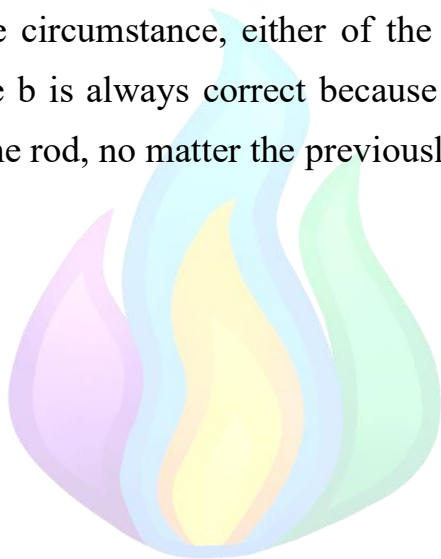
The only equation that follows the above condition is choice D.

Correct Answer: D

Question 47

Since the rod is metal, it will conduct heat. It is not clear enough what would be the strongest source of heat, conduction or convection, because convection is dependent on the size of the fire and the distance at which we are observing it, while conduction is dependent on the exact type of metal the rod is made from and the length of the rod. Since we are sure there is both conduction and convection, answer choice c is wrong. Depending on the circumstance, either of the statements a or d could be correct but answer choice b is always correct because the conduction will always heat up the other end of the rod, no matter the previously discussed circumstances.

Correct Answer: B



Question 48

Using the right-hand rule, we see that if we change the current's direction, the force will also change its direction accordingly, so answer choice A is incorrect. If we interchanged the magnet poles, that would swap the direction of the magnetic field, which would also, according to the right-hand rule, change the direction of the force, so choice B is incorrect. Using the right-hand rule, we see that the force would not change direction if both the currents and the magnetic field's direction got swapped; answer choice c is incorrect. We already saw from the past answers that changing the direction of only one of the magnetic field or the current changes the direction of the force, but now if we change the directions of both the magnetic field and current, then this is the correct answer.

Correct Answer: D

Question 49

This is a simple trigonometry equation.

$$\tan 30 = \frac{\text{opposite side}}{\text{adjacent side}} = \frac{\text{Height of the tree}}{\text{Length of the shadow}} = \frac{\text{Height of the tree}}{50\text{m}}$$

Height of the tree = 50 m * tan 30.

Correct Answer: C

Question 50

To solve this problem, we will calculate the area of each shape.

$$\text{Area of the Semicircle} = \frac{1}{2}\pi r^2 = \frac{1}{2}\pi \left(\frac{14}{2}\right)^2 = 77\text{cm}^2$$

$$\text{Area of the quadrilateral} = \text{Width} * \text{Height} = 14\text{cm} * 12\text{cm} = 168\text{cm}^2$$

$$\text{Area of the the triangle} = \frac{1}{2} * \text{base} * \text{height} = \frac{1}{2} * 14\text{cm} * 12\text{cm} = 84\text{cm}^2$$

$$\text{Total Area} = 77\text{cm}^2 + 168\text{cm}^2 + 84\text{cm}^2 = 329\text{cm}^2$$

Correct Answer: B